# A Study of the States Kinetics in NBTI Degradation by Two-Stage NBTI Model Implementation

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***Abstract*— This paper presents a simulation framework for reliability analysis of PMOS devices in the TCAD Sentaurus environment. The degradation of parameter is based on the numerical solution for the two-stage NBTI model mechanism. We demonstrate and analyze the voltage degradation, $V_{th}$ of a high-k $HfO_2$ dielectric pMOSFET structure with effective oxide thickness (EOT) of 1.092 nm. After 1000s of stress, the threshold voltage shift of higher stress bias shows higher degradation (~0.07V) compared to the lower stress bias (~0.008V).**

***Keywords*— NBTI; two stage model; DC stress**

## I. Introduction

Negative bias temperature instability (NBTI) has become a critical reliability issue due to the vast development of semiconductor devices technology nodes. The device parameters such as threshold voltage, mobility and drain current will deteriorate during the course of operation and eventually degrade the circuit performance over time [1].

Many researchers have proposed series of NBTI models in order to explain the NBTI mechanism, such as two-stage model [2], [4],[5] and reaction-diffusion model [6],[7] Among the series of model that have been devised, the two-stage NBTI model shows an excellent alignment with experimental results in terms of explaining the behavior of the device [2], [11],[12].

In this paper, we will demonstrate and analyze the voltage degradation, $V_{th}$ of a high-k $HfO_2$ dielectric pMOSFET structure with effective oxide thickness (EOT) of 1.092 nm using two-stage NBTI model.

## II. Two-Stage NBTI Model

### A. Two-Stage NBTI Model Mechanism

The two-stage NBTI model is associated with two types of defect, which is known as recoverable defect and permanent defect [8]-[10]. Recoverable defect is a defect that can return to its original condition after stress is applied. Permanent defect is a fixed or slowly recoverable component of NBTI.

The two-stage NBTI model involves two stage processes, where the first stage describes hole-capture process into the deep near-interfacial states and the second stage involves the creation of permanent defects [2],[3]. The model is associated with the kinetics of four states, namely $S_1$, $S_2$, $S_3$ and $S_4$. State $S_1$ consists of Si-Si dimer precursor which is generally known as the pre-existing defect. State $S_2$ and $S_3$ consist of positively charged E' center and neutral E' center respectively. State $S_4$ consists of fixed positive charge and interface trap, $P_b$ which is known to be permanent defect.

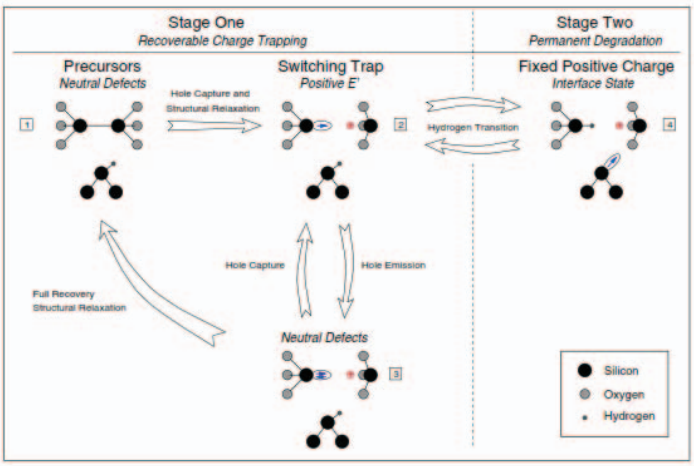


Fig. 1: The mechanism of the two-stage NBTI model [2]

In between the states, there are transitions from one state to another. Transition between states $S_1$ to $S_2$ occurs when the stress voltage is applied. The stress application will cause holes from the channel to obtain sufficient energy to tunnel into the precursor located near to the interface. The hole trapped in a precursor breaks up the bonds in the Si-Si dimer, causing a positively charged E' center and a silicon dangling bond created.

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The transition between state $S_2$ to $S_3$ and back to $S_2$ is during the emission and recapture of hole between state $S_2$ and $S_3$. The defect can be repeatedly charged and discharged by electrons tunneling in or out of its dangling bond.

From state $S_3$, the neutral E' center will recover the Si-Si bond in a period of time and return back to the original precursor, state $S_1$. The transition from $S_2$ to $S_4$ and back to $S_2$ embarks the second stage of the two-stage model [2]. In state $S_2$, there are a positively charged E' center and an unpassivated silicon dangling bond. This bond inside the *E'* center may become passivated by gaining a hydrogen which is removed from a passivated silicon dangling bond at the interface. The hydrogen locks in the positive charge and consequently delaying the recovery of the defect. The unpassivated dangling bond left by the hydrogen at the interface is known as interface trap ($P_b$) center.

## III. Device and Simulation Conditions

A high-k $HfO_2$ dielectric 32nm pMOSFET structure with effective oxide thickness (EOT) of 1.092 nm is simulated using Synopsys TCAD Sentaurus simulator [3].

## IV. Methodology

### A. Prestress Phase

The device is pre stress with DC bias voltage of -0.3V under the temperature of 400K. Initial graph of $I_d$-$V_g$ is obtained as shown in Fig. 2 in order to extract the fresh threshold voltage, $V_{tfresh}$ before the sample is exposed to electrical stress.

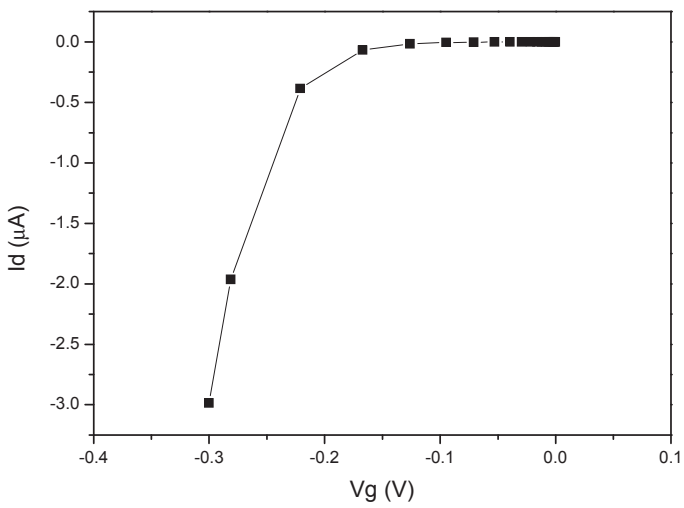


Fig. 2: Initial Id-Vg curve

### B. Stress Phase

Subsequently, stress voltage of -0.5V, -1.3V, -1.0V and -1.5V with oxide field (Eox) of -2.5MV/cm, -5.0MV/cm, -6.5MV/cm and -7.5MV/cm respectively is applied on the device with stress time of 1000s for each gate bias. The stress temperature is maintained at 400K. For different stress voltages, the drain current is monitored as a function of time. The relationship between the transconductance and the drain current change results in the extraction of the threshold voltage shift, $\Delta V_{th}$.

### C. Relaxation Phase

After each stress biasing, the device is relax by applying gate voltage of 0V for 1ks. The curve of threshold voltage shift, $\Delta V_{th}$ during relaxation is obtained for further analysis. The threshold voltage shift is affected by the presence of permanent and recoverable component. The recoverable component will return back to its original state after removal of stress voltage while the permanent component will not decrease or will be poorly recover after stress removal [8], [10], [14].

## V. Simulation Results and Discussion

### A. Threshold Voltage Degradation, $V_{th}$

Fig. 3 shows the threshold voltage degradation curve for different stress bias applied on the device and Fig. 4 shows the curve of relaxation of threshold voltage after stress bias is removed. The slope increases as higher stress voltage is applied. Gate voltage is often associated with carrier-energy driven degradation [9]. Higher oxide field due to high stress voltage causes more holes from the channel to obtain sufficient energy to tunnel into the precursor located near to the interface and hence, contribute to the degradation of the device.

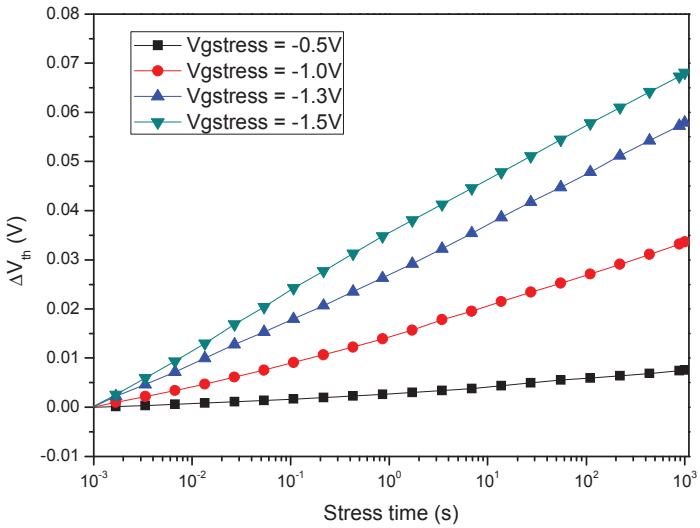


Fig. 3: Threshold voltage degradation for different stress bias

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The asymmetry between stress and relaxation curves is properly explained by the two-stage NBTI model. The relaxation phase, in which the phenomena of detrapping of charges occur, intervals over a much larger period of time compared to the stress phase (trapping of charges). The threshold voltage after stress voltage is removed does not totally return to its original level due to the generation of permanent defects [8], [10], [14].

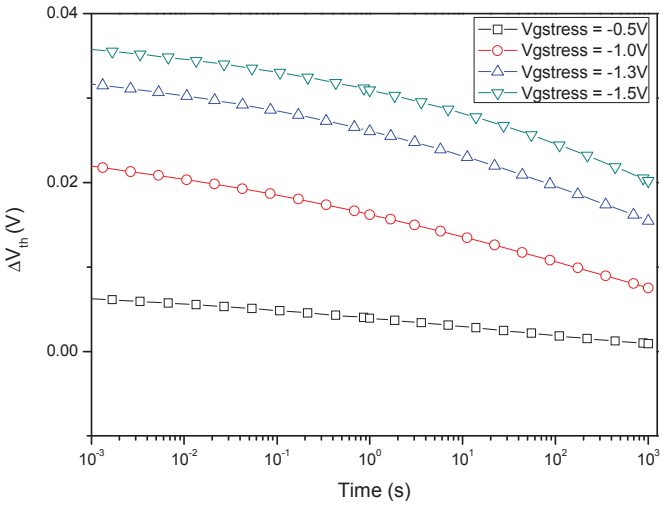


Fig. 4: Relaxation of threshold voltage after stress voltage is removed

### B. *Interface Trap, $N_{it}$*

Interface trap, $N_{it}$ is a defect located at the oxide/semiconductor interface in which the action of trapping and de-trapping charge carriers occur. Fig. 5 shows the interface trap generation, $\Delta N_{it}$ ($cm^{-2}$) for different stress bias applied on the device and Fig. 6 shows the interface trap after stress voltage is removed. The change in the average interface trap, $\Delta N_{it}$ clearly shows an increase in the interface state as the NBTI stress progresses and slowly recovers during the relaxation phase.

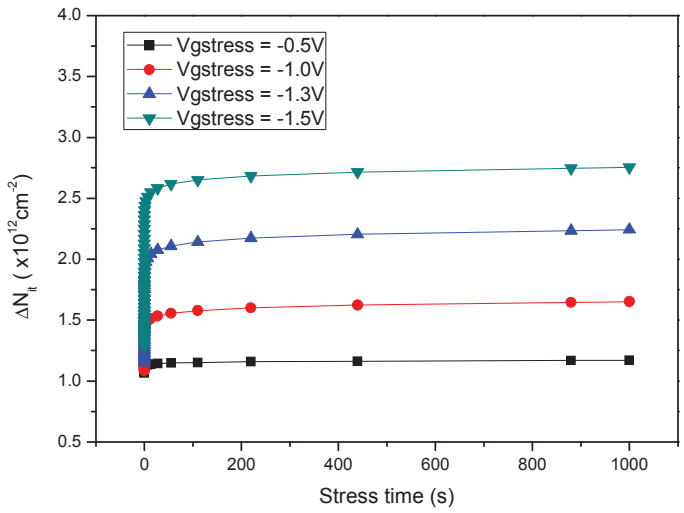


Fig. 5: Interface trap for different stress bias

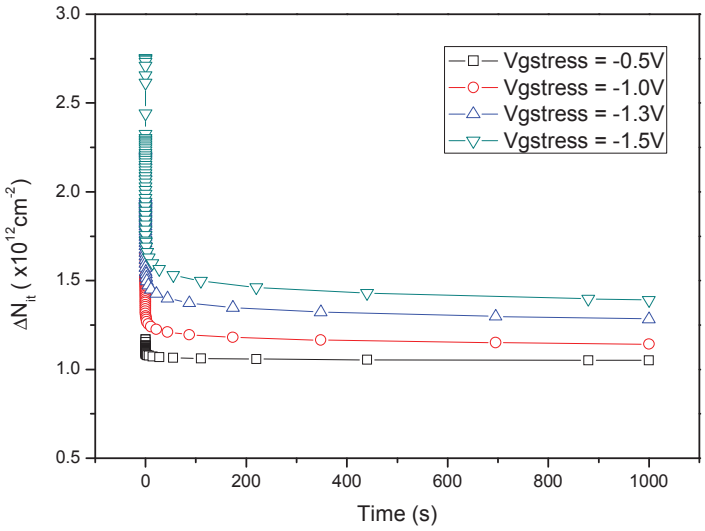


Fig. 6: Interface trap after stress voltage is removed

## VI. Conclusion

In conclusion, the two-stage NBTI model is able to explain the behavior of the device excellently in which is proven from the agreement between the model and simulation results. The two-stage NBTI model involves the process of two stages which takes the permanent and recoverable defects into account. The threshold voltage after stress bias is removed is unable to entirely return to its original level due to the generation of permanent defects.

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